

PROJECT REPORT

Ray-based Multi-Agent AI System for Stock Analysis

Ramakrishna Mission Vivekananda Educational and Research Institute

Submitted By

Tanmoy Paul

Sovan Pal

M.Sc Data Science and Artificial Intelligence

Academic Session: 2025–2026

Department of Computer Science

Abstract

Financial market analysis is challenging because stock movement depends on both historical market trends and external information such as financial news and investor sentiment. Traditional forecasting systems often focus only on numerical data and ignore contextual market information.

This project proposes a distributed multi-agent stock analysis framework using Ray, Time-VLM, FinBERT, FastAPI, and LLM-based reasoning. The system executes multiple agents in parallel, where FinBERT performs financial sentiment analysis and Time-VLM forecasts stock movement using multimodal time-series understanding.

The generated outputs are aggregated to produce an investment recommendation in the form of **BUY, SELL, or HOLD**, together with confidence score, risk estimation, future trend, and market insights.

Keywords: Ray, Multi-Agent System, FinBERT, Time-VLM, Stock Forecasting, Distributed AI

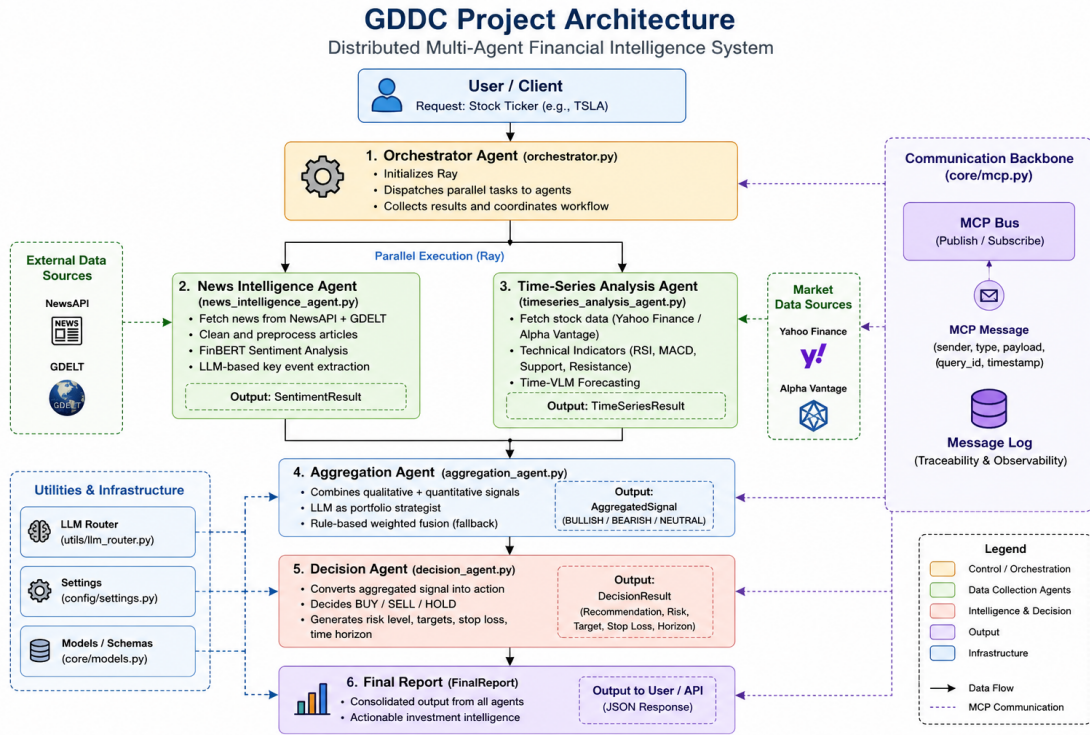
1 Problem Statement

Stock market prediction is a complex task because market movement depends on both historical price trends and external factors such as financial news and investor sentiment. Traditional analysis systems often process these sources separately, which increases computation time and reduces decision efficiency. Additionally, modern AI models require significant processing resources, making real-time analysis difficult.

This project addresses these challenges by developing a Ray-based distributed multi-agent AI system that performs parallel analysis of stock data and financial news. The system combines FinBERT for sentiment analysis and Time-VLM for time-series forecasting to generate faster and more informed investment decisions. The final output provides a clear BUY, SELL, or HOLD recommendation with improved scalability and reduced latency.

2 System Architecture

The proposed system follows a distributed multi-agent architecture where different agents execute in parallel through Ray. Each agent performs a specialized task and contributes to the final investment recommendation.



1. Orchestrator Agent

The Orchestrator Agent acts as the central controller of the entire system. It initializes the Ray distributed environment and coordinates communication between all agents. After receiving a user request containing the stock ticker, it dispatches tasks to different worker agents in parallel and collects their outputs. It then forwards the processed information to the aggregation and decision stages to generate the final report.

- Initializes Ray cluster
- Manages parallel execution
- Coordinates communication between agents
- Collects and returns final output

2. News Intelligence Agent

The News Intelligence Agent is responsible for analyzing market sentiment from financial news sources. It collects recent articles using external APIs and processes textual information using FinBERT. In addition, it extracts major financial events that may influence stock movement and converts them into structured sentiment information.

- Fetches financial news
- Performs FinBERT sentiment analysis
- Identifies key market events
- Produces sentiment score and market context

3. Time-Series Analysis Agent

The Time-Series Analysis Agent performs quantitative stock analysis using historical price data. It gathers stock market information and calculates technical indicators such as RSI, MACD, and volatility. The processed data is then passed through the Time-VLM model to forecast future market trends and stock movement.

- Collects historical stock data
- Computes technical indicators
- Executes Time-VLM forecasting
- Produces trend and prediction outputs

4. Aggregation Agent

The Aggregation Agent combines outputs generated by the News Intelligence Agent and Time-Series Analysis Agent. Using LLM-based reasoning, it fuses qualitative and quantitative signals into a single market understanding and calculates an overall confidence score.

- Combines sentiment and technical signals
- Performs multimodal reasoning
- Generates confidence score
- Creates unified market signal

5. Decision Agent

The Decision Agent is the final decision-making component of the architecture. It receives the aggregated market signal and generates an interpretable investment recommendation. Along with BUY, SELL, or HOLD suggestions, it also estimates risk level, target price, and investment insights.

- Produces BUY / SELL / HOLD recommendation
- Estimates target price
- Calculates risk level
- Generates final investment insight

3 Technology Stack

Component	Technology
Backend API	FastAPI
Distributed Computing	Ray
Sentiment Analysis	FinBERT
Forecasting Model	Time-VLM
Vision-Language Model	CLIP
LLM Reasoning	Groq + LLaMA-3
Data Sources	Yahoo Finance, NewsAPI, GDELT
Programming Language	Python

4 Ray Distributed Framework

Ray is used as the distributed computing framework to enable parallel execution of multiple agents. The system launches independent worker processes for news analysis and time-series forecasting,

- News Worker
- Time-Series Worker

allowing both tasks to run simultaneously.

This architecture reduces execution time, improves resource utilization, and enables scalable deployment. Ray actors also maintain model states across requests, minimizing repeated model loading overhead and improving overall system performance.

5 FinBERT Sentiment Analysis

FinBERT is a financial-domain transformer model used to understand sentiment from market news and financial text. It helps identify whether market conditions indicate positive, neutral, or negative investor sentiment.

Workflow:

- Collect financial news
- Preprocess and tokenize text
- Perform sentiment classification using FinBERT
- Generate sentiment score

Output: $\text{SentimentScore} \in [-1, 1]$

A positive score indicates bullish market sentiment, while a negative score indicates bearish sentiment.

6 Time-VLM Model

Time-VLM is a multimodal forecasting model used to analyze stock market time-series data by combining temporal, visual, and textual information. Instead of relying only on historical prices, it learns market patterns from multiple representations to improve forecasting performance.

RAL – Retrieval Augmented Learner

RAL focuses on learning temporal market behavior from historical stock data.

- Splits time-series into patches
- Retrieves similar historical patterns
- Learns temporal relationships for forecasting

VAL – Vision Augmented Learner

VAL transforms stock data into image representations and extracts visual information.

- Converts time-series into image format
- Uses FFT and periodic encoding
- Extracts visual features through CLIP

TAL – Text Augmented Learner

TAL adds textual understanding to improve prediction quality.

- Generates statistical text prompts
- Encodes market-related context
- Produces text embeddings

Fusion Layer

The final layer combines outputs from all three learners using cross-attention to generate predictions.

$$Prediction = Fusion(RAL + VAL + TAL)$$

Output:

- Forecasted stock trend
- Future price estimation
- Market direction analysis

7 Working Process

The system follows a distributed multi-agent workflow where different agents execute in parallel using Ray to reduce processing time and improve scalability.

1. The user provides a stock ticker as input.
2. The **Orchestrator Agent** initializes the Ray framework and distributes tasks to worker agents for parallel execution.
3. The **News Intelligence Agent** collects financial news and applies the **FinBERT** model to perform sentiment analysis and identify important market events.
4. Simultaneously, the **Time-Series Analysis Agent** downloads historical stock data, computes technical indicators, and executes the **Time-VLM** model to forecast future market trends.
5. The **Aggregation Agent** combines sentiment signals and forecasting outputs to generate a unified market understanding.
6. Finally, the **Decision Agent** produces the investment recommendation in the form of **BUY, SELL, or HOLD**, along with confidence and market insights.

8 Conclusion

This project demonstrates an intelligent financial analysis platform using distributed AI architecture.

By combining Ray for parallel processing, FinBERT for financial sentiment analysis, and Time-VLM for multimodal forecasting, the system produces faster and more informative market decisions than traditional sequential pipelines.

The architecture supports scalability, modularity, and future expansion toward real-time portfolio optimization and automated financial intelligence systems.

References

1. Ray Documentation
2. FinBERT Financial Sentiment Model
3. Time-VLM ICML 2025
4. FastAPI Documentation
5. CLIP Model Documentation